

TRIN ETRA

An AI that argues — under rules you wrote.

AI for sense-making and constitutional auditing · deterministic logic for scoring · humans for final approval. A live model makes the call; the record replays byte-identically.

4:07

TRINETRA

VS

~~6-DAYS~~

INDIA · MARCH 2026

PROPOSE

an AI suggests moves

ENFORCE

a machine—not a model—checks the rules

RECORD

every step hash-chained, offline

In March 2026, India needed six days to answer one question.

Not a data problem — a deliberation problem. The buffer isn't barrels; it's decision speed.

\$100M+

₹850+ cr — cost of the six-day decision lag

price path R · spot share E

\$270M

one SPR day burned at the peak, refilled later

arithmetic R · attribution E

\$30–45M

one VLCC bought in panic instead of on time

R — from the tape

Base inputs (R): imports ~4.9 mb/d (PPAC); Brent \$71→\$77→\$100+→\$126 (EIA). The app computes this live and stamps it “order-of-magnitude.”

“My comfort point used to be 60-60-60 — 60 days of crude, gas and LPG.”

— Hardeep Singh Puri, Union Minister for Petroleum & Natural Gas, 2026

R

WHAT IT COSTS

\$150K / desk / year

Breakeven \$0.0003/bbl · ROI ~200x vs one \$30M cargo

The ask: one design-partner desk — not a cheque.

Targets: IOC Intl Trade Directorate · RIL Jamnagar crude desk · BPCL risk cell

COMPUTED LIVE →

90 kb/d

REMAINING SHORTFALL

Started at 1658 kb/d · 95% firmly covered by 4 levers · 3260 kb/d more pending conditions

\$140M

COST OF A 6-DAY DECISION LAG

at the current price excess · every day the

A crisis room that convenes itself — and argues under your rules.

TRIN△ETRA

CRUDE DECISION GO

▶ DEMO

WHAT-IF

BACKTEST

When Hormuz closed, India took **6 days** to reroute crude. TRINETRA does it in **4 minutes** — and every rejection is a machine-checked rule, not an AI guess.

10 d

NATIONAL COVER · SAFE LINE 10 D

Day 65

PIPELINE TIME (REPLAYED FEED)
05:20Z · 10:50 IST

0:04

REAL TIME · CLOCK IS REAL

▶ Supply is disrupted — a shortfall of 1658 kb/d has opened. National days-of-cover is 10 (safe line 10).

The AI proposed 26 substitute moves. The rules-only critic approved 17, demoted 6, and blocked 3 — no AI in that decision.

■ approved / ok

■ conditional / stressed

■ blocked / critical

■ synthetic data

CACHED 2026-03-01

AIS AIS: anchorage queue at Fujairah swells past 60 vessels awaiting orders 15:00Z · 20:30 IST

MILESTONE OFAC sets loading cutoff date for Russia-related waiver eligibility 20:00Z · 01:30 IST

UKMT0 JMIC Adv

THE SHOCK

What's disrupting India's crude supply right now.

Hormuz severely disrupted

Hormuz partially disrupted + Hormuz severely disrupted

CLOSEST PAST CRISES · similarity computed from real facts

Iran–Iraq Tanker War 1984–88 · 0.5 mb/d, Brent +3% 71%

Red Sea / Houthi crisis 2023–24 · 4.7 mb/d, Brent +14% 39%

Ever Given / Suez blockage 2021 · 4 mb/d, Brent +6.4% 38%

THE DEBATE

An AI proposes moves. A rules-only critic (no AI) objects. An arbiter decides.

PROPOSER — the AI suggests substitute cargoes

demand cut

divert on-water cargo → Kochi ×14

floating storage

reroute spot cargo · Bonny Light (Nigeria) → Mangalore

reroute spot cargo · Cabinda (Angola) → Jamnagar

reroute spot cargo · Girassol (Angola) → Jamnagar

reroute spot cargo · Merey 16 (Venezuela) → Jamnagar

reroute spot cargo · Tupi (Brazil) → Jamnagar

reroute spot cargo · Urals (Russia) → Jamnagar

reroute spot cargo · Urals (Russia) → Mumbai

90 kb/d

REMAINING SHORTFALL

Started at 1658 kb/d · 95% firmly covered by 4 levers · 3260 kb/d more pending conditions

\$140M

COST OF A 6-DAY DECISION LAG

at the current price excess · every day the desk waits

STRAIT OF HORMUZ

+

−

WORLD ✓

FULL

THE RULES (CHARTER)

The constitution the AI argues under. Edit a number → the plan re-decides itself.

A1

Completeness

A2

Security floor

MIN DAYS OF COVER

10

A3

Concentration cap

MAX SHARE VIA ONE CHOKEPOINT

0.3

A4

Sanctions transparency

A5

Stated uncertainty

A6

Reversibility

A7

Traceability and human sovereignty

THE PROOF

CACHED

PRINT

Every step is hash-chained and re-runs byte-identically — offline.

20_arbiter_arbiter_ruled11a46hd0

THE PLAN

Top substitute cargoes the desk can execute now. Click one to trace its route on the map.

Cut demand / import products

APPROVED

Payment clear

arrives in 7 d

Execute 300 kb/d demand cut paired with product-import swap for 30 days. Net deficit coverage: 9 MMbbl. Cost: \$420M premium on swapped cargoes; GDP drag estimated at 0.15%. Binding constraint: A3_

CACHED AI RATIONALE · GENERATED OFFLINE AT BUILD TIME

Divert on-water cargo → Kochi

APPROVED

Payment clear

arrives in 3 d

Divert 2.1Mb owned on-water crude to Kochi (BPCL). Baseline cover jumps from 11 to 18 days immediately. Cost: \$4.2M diversion fees + \$0.85/bbl demurrage; total exposure capped at \$6.0M. Binding constraint: Single_

CACHED AI RATIONALE · GENERATED OFFLINE AT BUILD TIME

National shortfall 1658 kb/d — 34% of India's daily crude runs

strategic reserve draw 1000 kb/d · divert on-water cargo 201 kb/d · floating storage 67 kb/d · demand cut 300 kb/d · demoted pending conditions 3260 kb/d

One screen, one engine: signals → exposure-graph re-score → machine-generated option cards → charter debate → hash-chained audit. Live LLM at the edges; deterministic core runs offline.

△ TRINETRA

It demoted Venezuela's Merey. It didn't have an opinion — it ran the rules.

Article A2 · Security floor — voyage longer than days-of-cover

43 d voyage vs 12.2 d cover buffer — Article A2

FLAGS · cited: sc:venezuela-merey-q2, edge:venezuela-jamnagar

Article A4 · Sanctions — payment rail needs clearance

payment rail AMBER — OFAC-licensed only: GL 50A (13 Feb 2026) authorizes named majors + PdVSA dealings; per-cargo comfort letter. Snapshot 2026-03-01. — Article A4

FLAGS · cited: sc:venezuela-merey-q2, edge:venezuela-jamnagar

Article A5 · Refinery chemistry — crude too heavy/sour to run neat

grade Merey 16 (Venezuela) needs BLEND at Jamnagar: api 16 < min 18 · max_blend_ratio 0.87 binding — Article A5

FLAGS · cited: sc:venezuela-merey-q2, edge:venezuela-jamnagar

Three cited objections, computed by pure functions:

A2 43-day voyage vs a 12.2-day cover buffer.

A4 OFAC payment rail AMBER (snapshot 2026-03-01).

A5 Too heavy to run neat — API 16 < Jamnagar's min 18.

VERDICT **CONDITIONAL** (3 flags, 0 blocks)

Each objection cites its Article + evidence. No model can add, soften, or remove one.

“It ran the sanctions list, the assay table, and the calendar. The language model just wrote the memo.”

We replayed real crises — and every number shows how certain it is.

BACKTEST — real feeds, time-compressed

Red Sea / Houthi 2023-24 8.7 → 4.0 mb/d · Brent +14% lead time: +26 d early	R
Hormuz 2026 (live replay) ~150 tankers anchored · Brent >50% lead time: +4 d	R

“India's desks validated it: non-Hormuz sourcing rose 55% → 70%; suppliers widened 27 → 41 countries.” We didn't ask an expert — we let the market grade it.

EVERY CARD CARRIES A PROVENANCE CHIP

THE PLAN
Top substitute cargoes the desk can execute now. Click one to trace its route on the map. Cut demand / import products ● Payment clear arrives in 7 d Execute 300 kb/d demand cut paired with product-import swap for 30 days. Net deficit coverage: 9 MMbbl. Cost: \$420M premium on swapped cargoes; GDP drag estimated at 0.15%. Binding constraint: A3... CACHED AI RATIONALE · GENERATED OFFLINE AT BUILD TIME Divert on-water cargo → Kochi ● Payment clear arrives in 3 d Divert 2.1Mb owned on-water crude to Kochi (BPCL). Baseline cover jumps from 11 to 18 days immediately. Cost: \$4.2M diversion fees + \$0.85/bbl demurrage; total exposure capped at \$6.0M. Binding constraint: Single-... CACHED AI RATIONALE · GENERATED OFFLINE AT BUILD TIME

LIVE R real CACHED E estimated SYNTH S synthetic

The LLM runs LIVE (NVIDIA, via a key-holding edge worker) and every completion is recorded for replay; airplane mode falls back to the build-time cache. Per-refinery inventories are modeled (S), and we say so.

CHANGE ONE RULE → THE PLAN RE-DECIDES ITSELF

Raise the security floor 10 → 15. Long-haul cargoes fail — each cites the rule.

FLOOR 10 — CONDITIONAL

Article A2 · Security floor — voyage longer than days-of-cover

43 d voyage vs 12.2 d cover buffer — Article A2
FLAGS · cited: sc:venezuela-merey-q2, edge:venezuela-jamnagar

FLOOR 15 — REJECTED

Article A2 · Security floor — voyage longer than days-of-cover

43 d voyage vs 12.2 d cover buffer — Article A2
BLOCKS · cited: sc:venezuela-merey-q2, edge:venezuela-jamnagar

Same fact — 43-day voyage vs 12.2-day cover. The flag escalates to a block; the arbiter rejects it. Six Jamnagar-bound cargoes fall the same way.

THE RULES (CHARTER)

The constitution the AI argues under. Edit a number → the plan re-decides itself.

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Traceability and human sovereignty

THE PROOF

VERIFIED

PRINT

Every step is hash-chained and re-runs byte-identically — offline.

49_arbiter_arbiter ruled

be4bf93d

AND IT'S ON THE RECORD

Every verdict is hash-chained (seq · actor · input→output · prev) and re-runs byte-identically, offline. Revert to 10 and the earlier state's hashes reproduce exactly — no clock, no randomness on the derivation path. One click prints the trace on white: the signed prop.

“Run it again — byte-identical. Try that with a prompt.”

Boring, inspectable, event-driven.



solid = deterministic · proposer prose is LIVE (NVIDIA via a key-holding edge worker), recorded for replay; cache is the airplane-mode fallback

THE RIGHT TOOL PER LAYER — chosen by what each layer is allowed to get wrong

RULES	for the veto	must be citable & cannot hallucinate
OUR MODEL	for the number	softmax, frozen weights — byte-identical every run
ONE LLM	for the sentence	live at the edges, recorded for replay — only prose may vary

“One big LLM” would hallucinate the compliance check, vary the number every run, and be un-replayable — breaking the audit trail that is the moat.

Now it reads messy operator text — and the gate throws out whatever it makes up.



AI (blue) only proposes and observes. The green rail — validation + scoring — is deterministic and authoritative. A human signs.

THE GATE, LIVE — a hallucinated Brent 9000 and a made-up grade, rejected before scoring:

① AI SENSE-MAKING ADVISORY — LLM PROPOSES; RULES VALIDATE

shock: hormuz severe

~~*brent_usd = 9000~~

~~*charter.A2 = 200~~

🛑 2 candidate field(s) rejected by the validation gate — they never reached scoring.

THE MODEL, LIVE — NVIDIA, on every update:

Cut demand / import products APPROVED

● Payment clear arrives in 7 d

The desk would execute cutting demand to mitigate the crisis, as it has been validated by the rules engine. The key constraint is the national gap of 1658 kbd that needs to be addressed.

LIVE

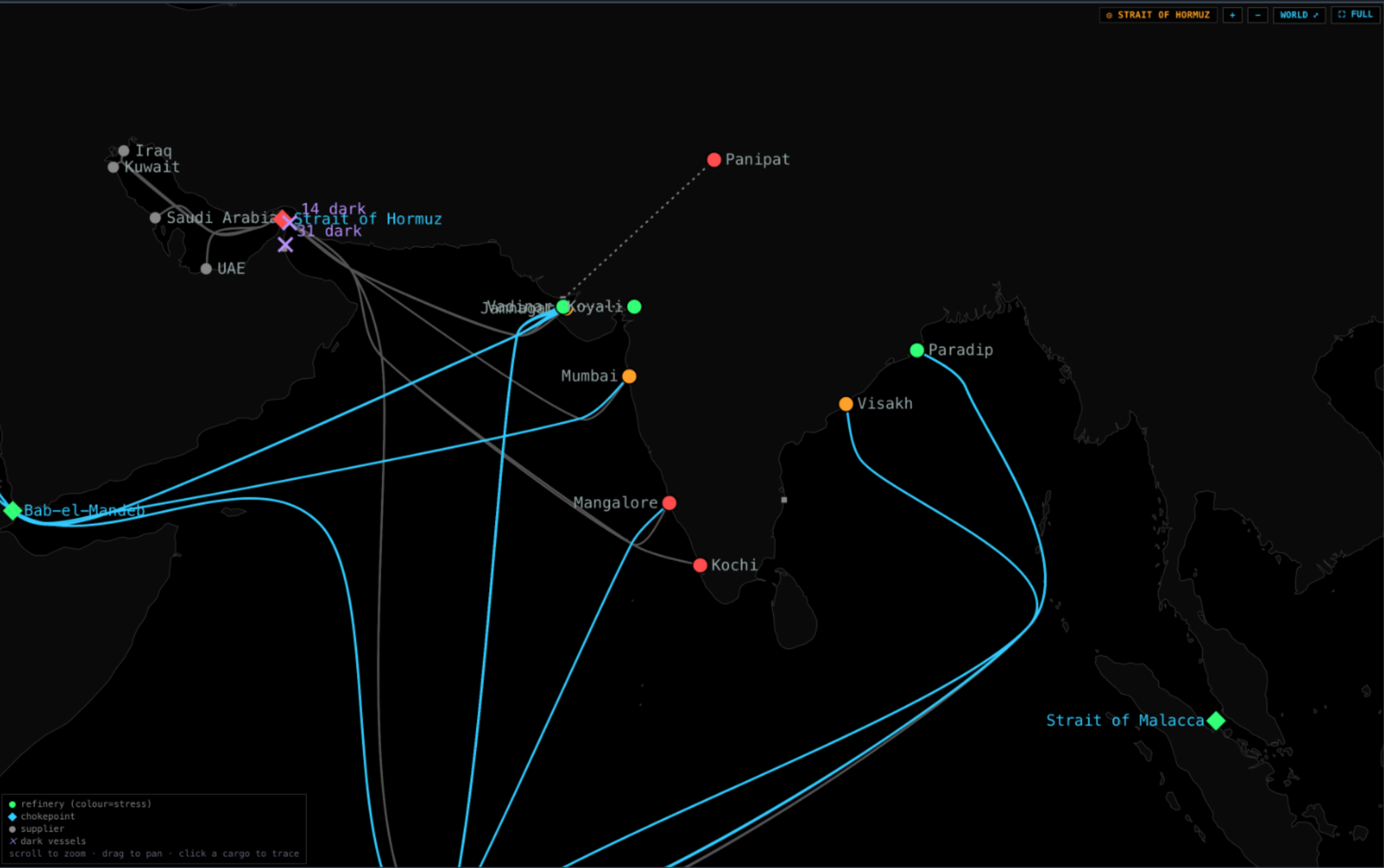
AI RATIONALE · META/LLAMA-3.3-70B-INSTRUCT · LIVE, RECORDED FOR REPLAY

live Llama-3.3-70B rationale, recorded for replay — captured from the app

A live model makes the call. The record replays byte-identically. A human signs.

The LLM runs LIVE on every decision-state change — NVIDIA Llama-3.3-70B through a key-holding Cloudflare Worker; the browser never sees a key. Every completion is RECORDED and replays byte-identically; the validation gate still drops anything fabricated before it can touch a score; scoring stays deterministic and offline-checkable.

One strait this week. Every chokepoint, every commodity, next.



The engine is graph-agnostic. A new theatre — a new chokepoint, a new commodity — is a new data bundle, not new code.

THE BUYER LADDER

- 01 Refiner trading desk
- 02 National oil company
- 03 Ministry crisis cell

Honest footer: live feeds (Kpler / Spire) are stubs behind the same 3-method FeedAdapter interface as the replay (src/engine/feeds.ts) — “production license.”

Six days to four minutes.

And the argument is on the record.

An AI that argues under rules you wrote. A machine — not a model — that enforces them. A decision you can audit.

DEMO + REPO github.com/banthia14aman/ET-Hackathon

TRIN△ETRA

THE RULES (CHARTER)

The constitution the AI argues under. Edit a number → the plan re-decides itself.

A1

Completeness

A2

Security floor

MIN DAYS OF COVER 15

A3

Concentration cap

MAX SHARE VIA ONE CHOKEPOINT 0.3!

A4

Sanctions transparency

A5

Stated uncertainty

A6

Reversibility

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Traceability and human sovereignty

THE PROOF ✓ VERIFIED ✓ PRINT

Every step is hash-chained and re-runs byte-identically – offline.

49_arbiter_arbiter_ruled_4e4bf93d

the signed audit trace — hash-chained, VERIFIED, one-click printable

“Did you already know the answer?” We built the firewall for that.

FROZEN KB

Structural knowledge cites only pre-2019 sources. The system doesn't “know about Angola” — it emerges from a query. Basrah Medium (a 2021 grade) was removed from the dataset entirely.

STATED CALIBRATION

Every calibration constant is a stated, sensitivity-swept assumption anchored to earlier crises (Abqaiq 2019, Suez 2021, Red Sea 2023-24) — never fit to the Hormuz-2026 outcomes the demo replays.

NO OUTCOME CLAIMS

We don't predict outcomes (nobody predicts a 30-day US waiver — that's an explicit branch). The testable claim: the options India chose were in our top five, surfaced weeks earlier.

MISSES SHOWN

It didn't score perfectly — the misses are on the scorecard, in red. Replayed real feeds, time-compressed; no live-feed claims. Scenario-choreography events are labeled SYNTH, never upgraded.